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Appendix	AI tool (e.g. ChatGPT)	How AI tool was used	Reference
A	ChatGPT 3.5	Asked for some ideas on how I might evaluate the usefulness of a piece of research	OpenAI. (2023). ChatGPT (Sept 12 version) [Large language model]. https://chat.openai.com/chat
B	ChatGPT 3.5	Asked for feedback on my paragraph structures	OpenAI. (2023). ChatGPT (Sept 14 version) [Large language model]. https://chat.openai.com/chat

Investigating Motor Learning in DCD: Precision and the Impact of Visual Cues

URN: 6741626

A dissertation submitted to the University of Surrey in partial fulfilment of the requirements
for the degree of BSc (Hons) in Psychology.

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Abstract:

Developmental Coordination Disorder (DCD) is a neurodevelopmental condition affecting individuals across multiple domains, with impairments most prominent in motor skills. Although motor deficits are well-documented in DCD, they are often studied in isolation, although over 50% of individuals with DCD also present with co-occurring neurodevelopmental conditions. This raises the question about how these co-occurrences may influence motor learning processes. To explore this, the current study examines how children with DCD perform in a visuomotor rotation task, an established method for assessing motor precision and learning, under varying conditions of visual feedback and learning type.

This study investigated whether children with DCD demonstrate reduced movement precision compared to typically developing peers and how visual feedback and practise affect performance. Six pilot participants completed an online visuomotor rotation task with both implicit and explicit learning conditions, as previous research suggests differing effects of DCD on these learning types.

Results indicate the design was effective, with significant changes across learning blocks. However, children with DCD did not consistently show reduced precision compared to neurotypical peers and showed similar performance in the two learning tasks. These findings contrast with much of the existing literature but align with recent work. However, due to high variability and a small sample size, the effect of co-occurring conditions was unable to be tested, and the results should be interpreted cautiously.

The success of the methodology supports its use in larger-scale future studies. With a target sample of 200 participants, future research could better capture group differences and explore how comorbidities affect motor learning in DCD.

Introduction:

Developmental Coordination Disorder (DCD) is a neurodevelopmental condition characterised by persistent motor coordination difficulties that impair daily functioning and persist throughout life (Zwicker et al., 2018). The Diagnostic and Statistical Manual of Mental Disorders, Fifth Edition (DSM-V; American Psychiatric Association, 2013) outlines four diagnostic criteria: Firstly, marked motor skill delays are inconsistent with age or experience; secondly, interference with daily activities or academic performance; the third states that exclusion of other medical or neurodevelopmental explanations is essential; and lastly, difficulties stem from adolescence and continue into adulthood. While "dyspraxia" is commonly used in the UK to refer to DCD, it lacks a consistent clinical definition and is not recognized in diagnostic systems like the DSM-V.

Children with DCD often experience delays in motor milestones and ongoing challenges with coordination, balance, and tasks such as running, catching, or handwriting (Gentle et al., 2024; Geuze, 2007). These persist, despite practice, reflecting atypical motor control systems (Schoemaker et al., 2015). Wilson et al. (2017) suggested that early identification has been found to be essential, as difficulties typically continue from adolescence into adulthood (Kirby et al., 2011), therefore limiting the effects from an early age is necessary.

Beyond motor challenges, children with DCD often face secondary issues such as academic struggles (Zwicker et al., 2012), social and emotional difficulties (Sylvestre et al., 2013; Harrowell et al., 2017), sensory processing issues (Tran et al., 2022), and reduced spatial navigation (Gentle et al., 2024). DCD has also been linked to poorer math performance (Pieters et al., 2012). These secondary issues can present the widespread effects that DCD can have on an individual. In spite of these challenges, some research highlights

compensatory strengths, such as resilience, creativity, and emotional insight (Maw et al., 2024). These may stem from the ways individuals with DCD engage with learning, particularly regarding implicit and explicit learning. Implicit learning is defined as non-episodic learning of complex information without awareness of what has been learned (Seger et al., 1994), whereas explicit learning is marked by consciously attended task demands and awareness of learning occurring (DeKeyser, 2003). Research supports that children with DCD may benefit more from explicit strategies, as their implicit learning appears less developed (Abswoude, 2019). Therefore, understanding these differences is key for designing effective educational and therapeutic strategies.

The prevalence of DCD is estimated at 5–6% of the population (Blank et al., 2019), though figures vary from 2–20% based on assessment criteria (American Psychiatric Association, 2013). Despite its comparable prevalence to conditions like autism spectrum disorder (ASD), attention-deficit/hyperactivity disorder (ADHD), and dyslexia, DCD is underrepresented in research (Biotteau et al., 2016). For instance, Maenner et al. (2023) estimated the ASD prevalence at 0.3%, Wagner (2020) reported 3–7% for dyslexia, and Cortese et al. (2023) found a 5.41% prevalence for ADHD. However, Biotteau et al. (2016) reports how all three of these are researched and understood more than DCD.

In addition to its underrepresentation in research, comorbidity is common in DCD, with over 50% of individuals with DCD having co-occurring neurodevelopmental conditions, including ADHD (~50%), dyslexia/dyscalculia (~40%), and specific language impairments (~11%) (Lino & Chiefoo, 2022; Schoemaker et al., 2013). These comorbidities can exacerbate motor and self-regulation challenges, making their consideration essential in assessment and intervention planning (Lingam et al., 2010; Cloninger, 2006).

The high rate of comorbidity in DCD contributes to an increase in complexity in diagnosis, which is further compounded by the condition's heterogeneity (Lust et al., 2022). Some children show poor balance but preserved fine motor skills, and some present the opposite pattern; this variation in presentation makes the need for accurate diagnosis critical. Diagnosis typically involves interviews, observation, and multiple standardised tools. One of which, the Movement Assessment Battery for Children (MABC), is the gold standard. The MABC evaluates manual dexterity, balance, and ball skills (Brown et al., 2009) and is an in-person assessment, though when this is not feasible, parent-report measures such as the Developmental Coordination Disorder Questionnaire (DCDQ) are also widely used and reliable.

Difficulties in DCD have been found to potentially stem from impairments in internal models; this is the brain's ability to predict sensory consequences of movements and update motor commands accordingly (Wolpert et al., 1998). In typical development, these systems allow rapid error correction and adaptation (Albert et al., 2016; Krakauer et al., 2019). On the other hand, in DCD, these processes may be slower or less accurate, potentially due to deficits in sensory integration, feedback processing, or prediction updating (Wilson et al., 2017; Adams et al., 2014). Given the complexity of these systems, it remains unclear whether motor deficits in DCD are due to impaired automatic motor learning, reduced strategic compensation, or both.

Impaired internal model function has wide-reaching implications for daily life, with impairments ranging from tying shoelaces to navigating school environments (Scott et al., 2023), which has been found to lead to avoidance, reduced confidence, and social isolation (Chen et al., 2003). Hence, understanding how children with DCD adapt to motor challenges is critical for both theory and intervention (SMITS-ENGELSMAN et al., 2013).

A widely used paradigm for studying internal models is the visuomotor rotation (VMR) task, in which visual feedback of a participant's movement is altered, typically by rotating the cursor away from the actual hand trajectory. This discrepancy between intended movement and visual outcome forces the brain to adapt to minimise the error. These tasks are particularly valuable because they allow researchers to distinguish between implicit learning and explicit learning. With given continued practice, participants adapt their movements to compensate for the rotation, demonstrating the brain's ability to update internal models of sensorimotor control. This allows for clear aftereffects, which are seen when the rotation is removed, as participants often continue to produce the adjusted movements. This presents the learning that has occurred throughout the task. However, Kagerer et al. (2004) found that children with DCD failed to show typical aftereffects, while later research (Kagerer et al., 2006) showed that, given more trials and stronger feedback, adaptation did occur, though more slowly, by using a larger rotation and more trials. These findings suggest that while internal model updating is possible in DCD, it may require extended training or more salient errors to engage adaptive processes.

Further research into DCD has yielded mixed results. Gomez and Sirigu (2015) used a visually guided pointing task and reported that children with DCD exhibited discrepancies between actual and imagined movement times, suggesting deficits in motor imagery. While Knaier et al. (2023) found comparable adaptation but greater variability and slower movement in DCD. These discrepancies underscore the need for further research.

The VMR task was adapted from Tsay et al. (2021) and offers a robust, online-friendly method to study these implicit and explicit processes in children. By manipulating feedback conditions and rotation angles, this design isolates learning strategies while maintaining ecological validity.

Introduction to our study:

This study examines motor adaptation and internal model updating in children with DCD (aged 9-11), using a modified version of Tsay et al.'s (2021) online VMR paradigm. Participants completed tasks involving 15° or 60° cursor rotations under varying visual conditions, assessing both implicit and explicit learning.

Our *first research question* asks whether children with DCD adapt to visuomotor rotation less precisely and show reduced retention of motor adjustments compared to neurotypical peers. **Hypothesis 1** predicts that children with DCD will be less precise (higher mean delta angle error) than their neurotypical peers, regardless of practice levels, and present slower evidence of learning through the task. Performance differences will depend on the available visual information, showing clear differences between implicit and explicit trials.

Our *second research question* examines whether co-occurring neurodevelopmental conditions affect precision and learning retention. **Hypothesis 2** predicts that children with co-occurring neurodevelopmental difficulties will show decreased precision and reduced learning retention, especially in visually demanding conditions. This includes children with conditions such as ASD, ADHD, and dyslexia, which may further influence attention, sensory integration, and learning strategies (Goulardins et al., 2024).

By combining theories of motor learning and internal models, this study aims to advance understanding of sensorimotor adaptation in DCD and guide development of targeted interventions.

Methods:

Participants:

The full study aimed to recruit 200 children aged 9–11 years old. Due to limitations, only pilot data from six participants is presented. Recruitment occurred via personal and professional networks. To encourage participation, families entered a prize draw for one of six £40 Love2Shop vouchers. Participants were split into two groups: DCD (N=2) and neurotypical (N=4).

Children were classified as neurotypical if their parents reported no neurodevelopmental conditions and their DCDQ score was within the typical range. The DCD group included children with reported DCD (suspected or diagnosed) and a DCDQ score below the cutoff. No participants had reported intellectual disabilities or visual impairments. Co-occurring conditions (e.g., ASD, ADHD and dyslexia) were only permitted in the DCD group.

Ethical approval was obtained, with informed parental consent and child assent secured (Appendix 1 & 2). Descriptive data are provided in Tables 1 and 2; see Appendix 8 for a detailed version.

Table 1. Group average descriptive data.

Condition	Number of Ps.	Gender	Mean age ± SD	Mean DCDQ ± SD
DCD	2	2 boys	10.80 ± 1.56	24.50
Neurotypical	4	2 boys, 2 girls	10.80 ± 0.96	67.75

Table 2. Individual descriptive data

Subject.ID	Condition	Gender	Age	DCDQ score
HOME06ting	DCD	Male	9.7	17
HOME11MAAW	DCD	Male	11.9	32
HOME15ting	Neurotypical	Female	11.7	66
HOME26LEIE	Neurotypical	Female	10.7	65
HOME7JUON	Neurotypical	Male	9.5	75

HOME12NAAM	Neurotypical	Male	11.3	65
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Study design and procedure:

The study included two parts: parent questionnaires and the child's task. These did not need to occur on the same day. After consent was obtained, parents completed demographic and health questionnaires on Qualtrics, including the Pubertal Development Scale (PDS) (Carskadon et al., 1993) and DCDQ (Wilson et al., 2007; Appendix 3).

The DCDQ is a 15-item parent-report screening tool for identifying motor coordination difficulties in children aged 5 to 15 years. Items assess motor performance in everyday activities (e.g., writing, running, balance) relative to peers, with strong psychometric properties including high internal consistency (Cronbach's $\alpha > .90$; Wilson et al., 2009). Age-specific cutoff scores were used to classify participants: scores ≤ 46 for ages 5-7, ≤ 55 for ages 8-9, and ≤ 57 for ages 10-15 indicated risk of DCD.

The PDS is a validated parent-report tool measuring observable signs of pubertal development. It includes items on physical changes such as growth spurts, body hair, and skin changes, with gender-specific items (e.g., menstruation for girls; voice deepening for boys). The responses are rated on a four-point scale (1 = has not begun to 4 = completed), allowing classification into developmental stages: prepubertal, early pubertal, mid pubertal, late pubertal, and post pubertal. This is used as puberty can cause many changes to individuals with DCD and how it may present itself, therefore these scores were used to ensure no participants had completed puberty.

Once completed and provided with an interest sheet (Appendix 4) and participant information sheet (Appendix 5), children completed the task at home, supervised by a parent who helped with understanding instructions but not task performance. Regular breaks and

attentional checks were used to maintain engagement throughout, and the task was hosted on Google Firebase and completed on a home laptop connected to the internet in a quiet setting.

We adapted the online VMR from Tsay et al. (2021), incorporating elements of experiments 1 and 2 in a gamified format. Children "fed" two aliens positioned at 45° and 135° using cursor movements. One trial was the movement from the centre point to the target point, and two trials made up one cycle. The layout mirrored Tsay et al. (2021, p. 12): targets were 6 cm from the start location, the cursor was 0.3 cm in diameter, trials began with a start click followed by a 500 ms delay, and movements had to be initiated within 800 ms.

Each participant completed two tasks, implicit and explicit, separated by a short break. The implicit task included 2 practice cycles and 68 main cycles: baseline no feedback (7 cycles), veridical feedback (7), clamped feedback (44), and an aftereffect block (10). No rotation information was provided. After the break, the explicit task included 2 practice cycles and the same baseline and veridical feedback blocks, followed by a 60° rotated feedback block (44) and an aftereffect block (7).

The study was structured into four key blocks, each designed to measure different aspects of motor adaptation. The first, baseline (no feedback), served as a familiarisation phase where participants completed trials without any visual feedback. This was followed by the baseline veridical block, which involved unmanipulated feedback to capture participants' performance under optimal conditions; the mean delta angle from this block was subtracted from subsequent blocks to allow for direct comparison. The clamped feedback block introduced a manipulated feedback condition (either 15° or 60° deviation), making participants adapt their movements. The implicit block ignored the actual movement, allowing measurement of implicit learning without strategic compensation. Whereas, during the explicit task, participants were informed of the rotation and instructed to adjust their aim,

facilitating a comparison of learning mechanisms (Mazzoni et al., 2006). This block was split into early and late learning phases for analysis. Finally, the aftereffect block removed all manipulation, assessing how well participants had adapted during the clamped phase and how quickly they returned to a 0° deviation. Measuring differences across these blocks enabled analysis of learning, precision, and retention effects throughout the task.

Data preprocessing and statistical analysis:

The primary outcome measure was the mean delta angle between the target and the hand, reflecting movement accuracy. For each cycle, the hand angle for both trials was averaged and subtracted from the corresponding target angle to produce a mean delta angle score. As mentioned, the baseline veridical data was excluded from statistical analysis once subtracted from the subsequent groups.

Trials were excluded using the criteria outlined by Tsay et al. (2021): trials with movement initiation slower than 800 ms or trials with hand angles deviating more than 90° from the target. Additionally, following the exclusion criteria set out by Tsay et al. (2021), trials were flagged as outliers if their hand angle deviated by more than 3 standard deviations from a moving average of five surrounding trials, indicating lapses in attention or anticipatory responses. In total, 32/1466 trials (2.2%) were excluded for slow movements and 204/1466 trials (13.9%) for angle deviation outliers, leaving 1230/1466 (83.9%) trials for analysis. Despite some participants having no valid explicit trials (Appendix 9), their implicit data were retained due to small sample size.

To address *Research Question 1*, separate analyses were run for implicit and explicit tasks, with corrected mean delta angle as the dependent variable. Independent variables were condition (DCD vs. neurotypical) and block (e.g., baseline, rotated/clamped feedback, aftereffect). The analysis across both the implicit and explicit tasks focused on changes across

the different feedback types, with implicit learning focused on clamped feedback and explicit focused on rotated feedback.

Prior to inferential analyses, normality was assessed with the Shapiro-Wilk test. The implicit condition met parametric assumptions ($W = 0.963$, $p = .503$; Appendix 14); whereas, the explicit condition did not ($W = 0.890$, $p = .038$; Appendix 15). For the implicit data, Levene's test indicated that the assumption of homogeneity of variances was met ($F(7, 16) = 0.944$, $p = .501$), justifying the use of one-way ANOVA for group comparisons (Appendix 16). Despite the violated Shapiro-wilk test, a parametric ANOVA was used for the explicit data, as non-parametric alternatives were not feasible and Levene's test confirmed no significant violation of homogeneity of variances ($F(1,16) = 0.03$, $p = .859$) (Appendix 17). This supported the decision to proceed with a parametric approach.

Two correlation matrices (Appendix 20 & 22) examined relationships between DCDQ scores and performance across blocks for each group using corrected delta angle values. Post hoc comparisons with Bonferroni correction (Appendix 19) assessed group differences where appropriate.

Hypothesis 2 could not be tested due to challenges with data collection, and participant recruitment prevented us from reaching the target sample size of 200.

Ethical considerations:

Ethical approval was granted by the University of Surrey Ethics Committee (Appendix 6 & 7). Informed consent and assent were obtained from the participant and the parent/guardian (Appendix 1 & 2), and participants could withdraw at any time. All data was anonymized using self-generated subject ID codes, securely stored, and only accessible to the

research team. Any emails used for the prize draw were deleted post-study and anonymized data were uploaded to the Open Science Framework (OSF).

Results:

Descriptive plots:

Figure 1 – Implicit data scatterplot

Implicit data scatterplot

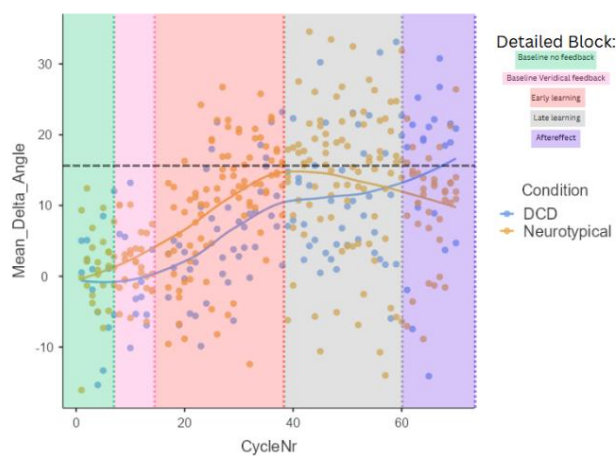


Figure 2 - Explicit data scatterplot

Explicit data scatterplot

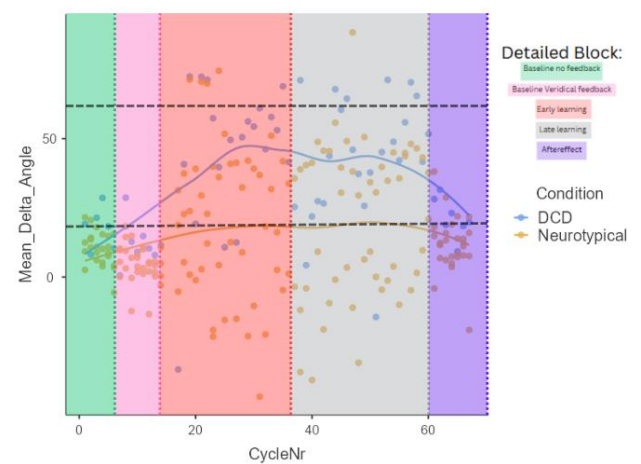


Figure 1 and Figure 2 (Appendix 10 & 11) display data for each cycle in the implicit and explicit learning tasks, respectively. In both tasks, the initial blocks served as baseline phases, where delta angle errors were expected to be close to zero, indicating accuracy prior to perturbation. Learning was indicated by gradual alignment with the rotation angle (15° or 60°), and retention was inferred from deviations in the final aftereffect block (Appendix 12 & 13). Values departing from zero in the aftereffect phase suggest successful internal model updating.

High variability was observed, particularly in the explicit task, likely reflecting individual differences and the small sample size. For example, the neurotypical group showed minimal adaptation in the explicit condition, though this is likely due to individual variability

rather than a true absence of learning, suggesting that these patterns should be interpreted with caution.

Implicit learning:

A one-way ANOVA (Appendix 18) revealed a significant main effect of block on delta angle difference ($F(3, 16) = 4.27, p = .021, \eta^2_p = .45$), indicating that performance varied across the learning phases. However, the main effect of condition was not significant ($F(1, 16) = 0.91, p = .353, \eta^2_p = .05$), neither was the Block $\times$ Condition interaction ($F(3, 16) = 0.52, p = .675, \eta^2_p = .09$).

Tukey's post hoc comparisons (Appendix 19) identified the significant difference between the baseline and aftereffect phases ($p = .026, d = -1.96$) and between baseline and late learning ($p = .040, d = -1.82$), supporting the effectiveness of the implicit learning paradigm. All other comparisons were nonsignificant ($ps > .27$).

Pearson correlation analyses (Appendix 20) revealed no significant associations between DCDQ scores and task performance: late learning ($r = .02, p = .970$), aftereffect ($r = -0.47, p = .347$), or baseline values ($r = -0.34, p = .509$). Baseline values were also not significantly related to late learning ($r = .40, p = .433$) or aftereffect ($r = -0.11, p = .840$). These results suggest that reported motor difficulties did not significantly correlate with implicit learning performance in this sample.

Explicit learning:

A 2 (Condition: DCD vs. Neurotypical) $\times$ 4 (Block: Baseline, Early Learning, Late Learning, Aftereffect) mixed-design ANOVA (Appendix 21) was conducted on corrected delta scores. No significant main effects or interactions emerged: Block ($F(3, 10) = 1.71, p = .227, \eta^2_p = .34$); Condition ($F(1, 10) = 0.20, p = .663, \eta^2_p = .02$); Block $\times$ Condition ($F(3, 10)$

= 0.15, $p = .931$, $\eta^2_p = .04$). Due to the absence of significant effects, no post hoc comparisons were conducted.

Spearman correlation analysis (Appendix 22) also showed no significant relationships between DCDQ scores and task performance: late learning ($\rho = -0.32$, $p = .233$), aftereffect ($\rho = -0.41$, $p = .090$), or baseline values ($\rho = 0.25$, $p = .317$). However, a significant positive correlation was found between baseline and aftereffect performance ($\rho = 0.80$, $p < .001$), suggesting that initial motor control may influence subsequent adaptation.

Overall, while the implicit learning condition demonstrated significant effects over time, no group differences were observed in either condition. The small sample size likely limited statistical power, and results should be interpreted cautiously.

Discussion:

Overview of Main findings

This study investigated motor learning in children with DCD using an online VMR task. Both DCD and neurotypical groups showed evidence of adaptation, particularly in the implicit learning condition. However, as seen across both scatterplots (Figures 1 & 2), high variability suggests that adaptation in DCD may be inconsistent and shaped by task demands.

Performance differences between implicit and explicit learning support previous findings (Abswoude, 2019). Implicit adaptation appeared largely preserved in DCD, while the explicit task, which required strategic re-aiming, showed greater individual error. This suggests that while error-based motor learning is intact, it may be more vulnerable under cognitive load.

These findings refine the internal model hypothesis in DCD. Rather than reflecting a complete deficit, motor learning in DCD appears fragile and context sensitive. Nevertheless, variability across tasks highlights the need for carefully designed interventions that account for individual learning patterns and task complexity.

The child-friendly, online VMR task (adapted from Tsay et al., 2021) also proved effective in eliciting reliable motor learning across both groups, demonstrating its utility for accessible research in diverse populations.

Preservation of implicit learning in DCD

Although no significant group differences were found, the implicit condition can still support preserved sensorimotor adaptation in children with DCD. Both groups showed varying levels of adaptation to a visuomotor perturbation without being informed of the rotation, suggesting intact implicit learning processes. The clamped feedback emphasises that adaptation was likely driven by sensory prediction error rather than strategic adjustments.

These findings align with theories that implicate the cerebellum and low-level feedback mechanisms in implicit learning, systems that may remain functional in DCD (Bianco et al., 2024). The ability of children with DCD to adapt comparably to their neurotypical peers, despite potential motor coordination challenges, suggests that fundamental motor learning mechanisms may be preserved. This supports earlier research indicating that children with DCD can show similar patterns of adaptation, though sometimes with reduced precision or slower rates (Knaier et al., 2023; Kagerer et al., 2006).

Difficulties with explicit learning: strategic load matters

As with the implicit condition, the explicit learning task revealed no statistically significant differences between groups. However, trends within the data suggest that some

children with DCD may have found it more challenging to consistently apply and sustain a re-aiming strategy in response to a 60° perturbation.

While not statistically significant, this observed variability within the DCD group may reflect difficulties with executive functioning, such as working memory and cognitive flexibility, that are commonly associated with DCD and related neurodevelopmental conditions (Lachambre et al., 2021). The explicit task imposed a higher cognitive load by requiring continuous, conscious adjustments based on feedback, which may have strained strategic consistency for some participants. In future research with a higher number of participants, these trends may present a statistically significant difference, as individual differences would play a smaller role.

These results reinforce the idea that task demands play a crucial role in motor learning outcomes. Even in the absence of group-level differences, explicit adaptation appears more sensitive to individual differences in cognitive resources, which may contribute to less stable performance in some children with DCD.

Internal model instability

According to internal model theory (Wolpert et al., 1998), motor learning relies on refining predictions through sensory error correction. In DCD, increased noise in sensory feedback and greater reliance on reactive corrections may lead to less stable predictive control and slower adaptation. This variability likely reflects a slower, less efficient learning process and difficulty generalising skills across contexts. The value of this stems from the potential explanation to the findings that were presented, as despite a lack of significant findings, the overall trend of the explicit data does support this idea. Furthermore, while this does not imply a failure to learn, it suggests learning may be less stable and more vulnerable to interference. Notably, a strong positive correlation between baseline and aftereffect

performance ($\rho = 0.800$, $p < .001$) emphasises how initial motor control ability influences the learning trajectory, underscoring the importance of individualised approaches in both theory and intervention.

Implications for practice

The trends that are present within the results support interventions that reduce cognitive load and leverage implicit learning in DCD. Rather than strategy-heavy instruction, therapies should focus on repetition, modelling, and minimal verbal guidance. The use of gamified or hands-on tasks promoting exploration may better support motor learning than activities requiring constant conscious correction. Furthermore, tools like clamped feedback systems, digital or physical, can improve sensorimotor control without overwhelming cognitive systems. This approach aligns with Schoemaker et al. (2015), who recommended low-load, practice-oriented environments that foster learning, reduce frustration, and enhance self-confidence. The incorporation of tasks with varying complexity levels, where children build on implicit learning before facing explicit demands, may promote more consistent motor development. The success of the online VMR task highlights the potential for remote tools in assessment and training, offering objective insights and supporting hybrid therapy approaches, especially in resource-limited settings.

Limitations and future directions

The study's small sample size ($N = 6$) limits generalizability and statistical power, but observed patterns align with existing research and warrant follow-up in larger samples to confirm group-level effects and examine individual differences. In addition to this, the absence of a delayed retention phase prevents conclusions about long-term learning. This indicates that future research should assess whether adapted responses are maintained over time and test long-term effects, possibly through a longitudinal study or repeated testing with

the same participants over an extended period. Not only that, but co-occurring conditions could not be adequately studied due to limited diagnostic data. Future work should replicate this design with a larger sample (target $N = 200$) to explore how conditions like ADHD, ASD, and others influence adaptation and strategy use in DCD.

The number of trials, especially in the aftereffect phase, may have been insufficient to fully capture learning, supporting the work of Kagerer et al. (2006). This is seen as the continued adaptation in the baseline no-feedback block of the explicit task suggests learning persisted beyond formal blocks, highlighting the need for more aftereffect trials in future designs. Lastly, the online format introduced variability in testing conditions (e.g., screen size, device type, and parental involvement) (Anwyl-Irvine et al., 2021). While standardisation and attentional checks helped mitigate these issues, they remain limitations, suggesting hybrid designs combining online tasks with lab-based controls may provide a more robust solution.

Conclusion

This study enhances our understanding of motor learning in children with DCD. The findings show that while adaptation to sensorimotor perturbations is possible, it is less efficient and precise, especially under conditions requiring cognitive strategies (explicit learning). Implicit learning mechanisms appear preserved, as significant differences between block presented learning across groups, suggesting that children with DCD maintain foundational motor learning abilities, similar to neurotypical individuals, that can be utilized in interventions. These results support the design of low-load, experience-based interventions focusing on implicit learning rather than explicit strategy-based methods.

The distinction is not between those who can and cannot learn but between those who learn consistently and those whose learning is more variable and context dependent. This

shift emphasises individual patterns of adaptation, supporting a strength-based approach. Instead of framing DCD as a complete disruption of learning, these findings suggest that adaptation can occur, though inconsistently and under certain conditions. This insight ultimately lays the foundation for personalised interventions that target specific needs, leading to more effective therapeutic approaches.

By aligning instructional strategies with preserved implicit learning capacities and reducing cognitive demands, educators and therapists can create more accessible environments for motor skill development in DCD. The success of the online VMR task demonstrates the potential of remote, scalable tools for assessment and training, particularly in resource-limited settings. However, the study's small sample size and inability to investigate co-occurring conditions highlight the need for larger, more comprehensive studies to understand how these factors impact learning outcomes. Additionally, exploring long-term retention through longitudinal studies or repeated testing will provide a fuller picture of motor learning in DCD. Ultimately, these findings support a shift toward individualized, flexible models of motor learning that emphasise reducing executive load to promote meaningful, sustainable adaptation.

Reflection

One of the main limitations of this dissertation was data collection. Although the original goal was to recruit 200 participants, only seven were collected due to time constraints, one of which was later excluded. This significantly limited statistical power and restricted the types of analysis that could be run. We recognised this early in March and adapted the project into using the pilot study data to assess the reliability of the experimental design for future research. If repeated, I would conduct a power analysis earlier and create a recruitment timeline to manage this process more effectively.

As a researcher, I demonstrated a high level of independence, especially when approaching the statistical analysis. Due to the nature of the data and not collecting enough data for variability, I revised my research questions and hypotheses and made informed choices about appropriate analyses, despite challenges posed by the sample size. Furthermore, managing feedback was essential; at times my understanding was incomplete, but I actively engaged with my supervisor's input to improve my work while filling in my gaps in knowledge.

Through this project, I have developed a range of valuable skills, from designing to deploying a dynamic Qualtrics questionnaire, and gained a stronger grasp of statistical reasoning and test justification. My ability to reflect and revise also improved significantly as I continuously refined sections for clarity, focus, and efficiency. The dissertation pushed me to think critically, write concisely, and make well-justified decisions at each stage of the research process.

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Acknowledgements

Group work contribution form:

Groupwork Contribution Template: PSY3065 - Dissertations conducted in a group

The final year dissertation aims to provide students with the opportunity to undertake an in-depth, empirical investigation of a psychological problem or topic. As part of this, students may be working in a group with others on the same research project. Most research is conducted in a group (or research team). As such, working on your dissertation project in a team mirrors what researchers do in their day-to-day jobs.

This document is intended to help your group to plan, monitor, and formally recognise the contributions of all group members to support effective group work. A statement of contributions is standard practice when submitting journal articles for publication.

You will create **one group contribution statement per group** – so all group members will agree on contributions. Each group member will need to upload the signed group contribution in their dissertation - after the Acknowledgements Section (see Dissertation Guidelines for more detail).

Planning: Early in the project you will need to use this to plan together who will contribute to each component/aspect of the task and distribute roles and responsibilities.

Monitoring: Keep track of group members' contributions together to ensure your group is progressing towards the goals/deadline. Amend the plan as needed (e.g., if someone is ill). If there are problems, it is **your responsibility** to try to resolve them as a group and to alert your dissertation supervisor to help resolve the situation, if necessary. Suggested deadline for completing the contribution statement is after data collection for the dissertation has been completed.

Submission: Please include the final signed contribution in your dissertation after the Acknowledgements Section of your dissertation. **All group members must sign this statement** using their signature or a personal but non-identifiable mark.

Note that:

- Not every group member is expected to contribute to every component. Effective teamwork involves distribution of tasks and working to each other's strengths. Discuss how you each prefer to work, your strengths, and any additional barriers or needs that you are comfortable sharing so that all group members can contribute and be valued.
- You must **coordinate as a group**; plan task allocation and deadlines; communicate and collaborate effectively; support each other; resolve disagreements constructively in a courteous and professional manner; embrace diverse skills, experiences, and perspectives. Plan how you will keep in touch and share work.
- The content of the group contribution statement will **not** affect your mark. However, this process and document should prompt group members to discuss difficulties or concerns.

Contribution Statement:

- In the table below, list all of the components of the task and group work involved in this assignment. These will vary by dissertation project. Here are a few examples of tasks: preparation (e.g., planning, researching, critical analysis), planning/experimental design, understanding of ethical considerations, creation of information sheet/consent form, submission of Ethics RM application form, creation of Qualtrics survey, creation of interview schedule, creation of experiment in Gorilla, creation of stimuli material, participant recruitment tasks, how many participants each member

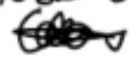
recruited/collected, and data collection. You can add components that arise later in the process.

You may find a shared document useful so that you can all access and edit it. Deciding how you go about this will be part of the group task, but you need to discuss and agree a final draft.

List the relevant tasks or responsibilities here	List the group members contributing to each component here. The submitted version should only list URNs to preserve anonymity.
Planned timeline with deadlines and meeting	6704390
Ethical considerations and ethics form	6741626 6752840
Created Qualtrics survey	6704390 6741626 6752840
If you have any additional comments about how your group worked in this assignment, please include them here.	
Due to time constraints, only pilot participants were recruited. This means each group member did not collect any data themselves, as we were not able to. For the Qualtrics, 6741626 completed the writing of questions, then both 6752840 and 6704390 completed the later stages and checked over all before completing and putting live for data collection.	

Signatures:

All group members must sign this group contribution statement. Please add your signature or URN.

Group Member URN	Signature
6741626	
6752840	
6704390	